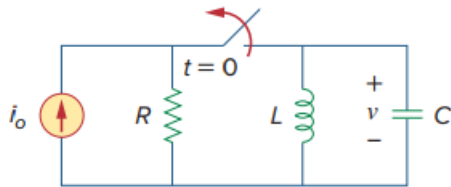


Problem

Find $v(t)$ for $t > 0$ in the circuit of Fig. 8.98.

Figure 8.98:



Step-by-step solution

Step 1 of 9

Refer to Figure 8.98 in the text book.

Initially at $t < 0$, the switch is closed for a long time. The circuit has reached to steady state and the inductor acts like a short circuit, while the capacitor acts like an open circuit.

The equivalent circuit diagram is shown in Figure 1.

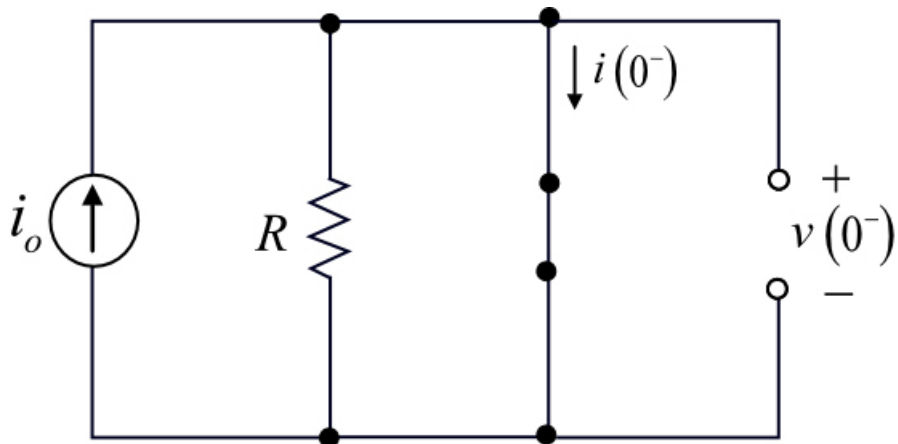


Figure 1

Step 2 of 9

In Figure 1, there is a short circuit across the resistor, R . Thus, no current flows through the resistor, R . Redraw the modified circuit diagram.

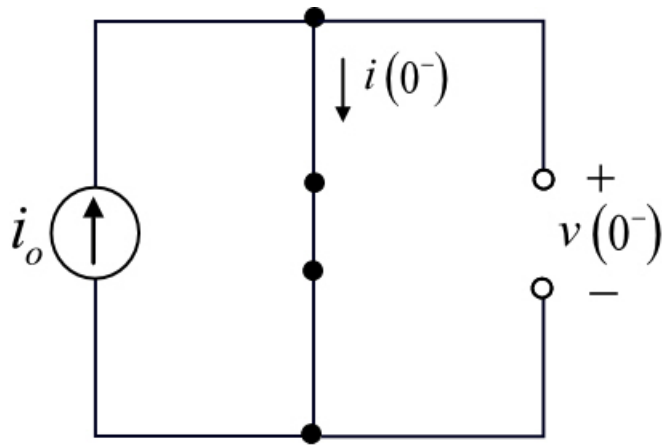


Figure 2

Step 3 of 9

From Figure 2, the initial voltage across capacitor is,

$$v(0^-) = 0 \text{ V}$$

The capacitor voltage does not change instantaneously.

$$\begin{aligned} v(0^+) &= v(0^-) \\ &= v(0) \\ &= 0 \text{ V} \end{aligned}$$

From Figure 2, the initial current through the inductor is,

$$i(0^-) = i_o \text{ A}$$

The inductor current does not change instantaneously.

$$\begin{aligned} i(0^+) &= i(0^-) \\ &= i(0) \\ &= i_o \end{aligned}$$

Step 4 of 9

For $t > 0$, the switch is opened. The equivalent circuit diagram is shown in Figure 3.

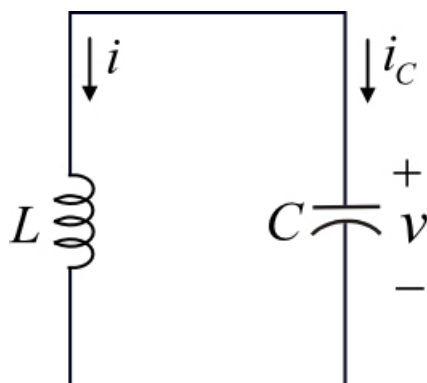


Figure 3

Step 5 of 9

From Figure 3, it is a source-free parallel LC circuit.

Write the formula for neper frequency:

$$\alpha = \frac{1}{2RC}$$

In Figure 3, there is no resistance in parallel with the energy storing elements.

Therefore, the value of neper frequency is,

$$\begin{aligned}\alpha &= \frac{1}{2(\infty)(C)} \\ &= 0 \text{ rad/s}\end{aligned}$$

Write the formula for resonance frequency:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

From the above calculations, $\alpha < \omega_0$.

Therefore, it is an under damped system.

Step 6 of 9

For an under damped system, the expression for voltage is,

$$v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

Here,

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Substitute 0 rad/s for α in the equation.

$$\begin{aligned}\omega_d &= \sqrt{\omega_0^2 - (0)^2} \\ &= \omega_0\end{aligned}$$

Therefore, the expression for voltage across the capacitor is,

$$v(t) = e^{-\alpha t} (A_1 \cos \omega_0 t + A_2 \sin \omega_0 t)$$

Substitute 0 rad/s for α in the equation.

$$v(t) = e^{-(0)t} (A_1 \cos \omega_0 t + A_2 \sin \omega_0 t)$$

$$v(t) = A_1 \cos \omega_0 t + A_2 \sin \omega_0 t \dots\dots (1)$$

Step 7 of 9

Determine the value of A_1 .

Substitute 0 for t in equation (1).

$$v(0) = A_1 \cos(\omega_0(0)) + A_2 \sin(\omega_0(0))$$

Substitute 0 V for $v(0)$ in the equation.

$$0 = A_1(1) + A_2(0)$$

$$A_1 = 0$$

From Figure 3, the current flowing through the capacitor is,

$$i_c = -i$$

Step 8 of 9

Differentiate the voltage expression with respect t , to determine the value of A_2 .

$$\begin{aligned}\frac{dv(t)}{dt} &= \frac{d(A_1 \cos \omega_0 t + A_2 \sin \omega_0 t)}{dt} \\ \frac{1}{C} i_C(t) &= \frac{d(A_1 \cos \omega_0 t + A_2 \sin \omega_0 t)}{dt} \quad \left(\text{since } v_C(t) = v(t) = \frac{1}{C} \int i_C(t) dt \right) \\ -\frac{1}{C} i(t) &= -A_1 \omega_0 \sin \omega_0 t + A_2 \omega_0 \cos \omega_0 t \quad (\text{since } i_C = -i)\end{aligned}$$

Substitute 0 for t in the equation.

$$\frac{1}{C} i(0) = -A_1 \omega_0 \sin(\omega_0(0)) + A_2 \omega_0 \cos(\omega_0(0))$$

Substitute i_o A for $i(0)$ in the equation.

$$\begin{aligned}-\frac{1}{C} i_o &= -A_1 \omega_0(0) + A_2 \omega_0(1) \\ -\frac{1}{C} i_o &= A_2 \omega_0 \\ A_2 &= -\left(\frac{i_o}{\omega_0 C} \right)\end{aligned}$$

Step 9 of 9

Recall the equation (1).

$$v(t) = A_1 \cos \omega_0 t + A_2 \sin \omega_0 t$$

Substitute 0 for A_1 and $-\left(\frac{i_o}{\omega_0 C}\right)$ for A_2 in the equation, to derive the expression for voltage across the capacitor.

$$\begin{aligned}v(t) &= (0) \cos \omega_0 t + \left(\frac{-i_o}{\omega_0 C} \right) \sin \omega_0 t \\ &= -\left(\frac{i_o}{\omega_0 C} \right) \sin \omega_0 t \text{ V}\end{aligned}$$

Thus, the expression for voltage across the capacitor is,

$$\boxed{v(t) = -\left(\frac{i_o}{\omega_0 C} \right) \sin(\omega_0 t) \text{ V}}$$